

C Programming – Summer Training

Assignment 5 Submission

Programs 1 – 8

Contents

- 1 Task 5.1 – WAP in C to find sum of digits of a number by using function
- 2 Task 5.2 – WAP in C to find factorial of a given number using function
- 3 Task 5.3 – WAP in C to find reverse of a given number by using function
- 4 Task 5.4 – WAP in C to check given character is vowel or consonant
- 5 Task 5.5 – WAP in C to change lower case characters of a string to upper case by using function
- 6 Task 5.6 – WAP in C to change upper case characters of a string to lower case by using function
- 7 Task 5.7 – WAP in C to find factorial of given number by using recursion
- 8 Task 5.8 – WAP in C to check given number is prime or not

Project 1 — Task 5.1

WAP in C to find sum of digits of a number by using function

Source Code

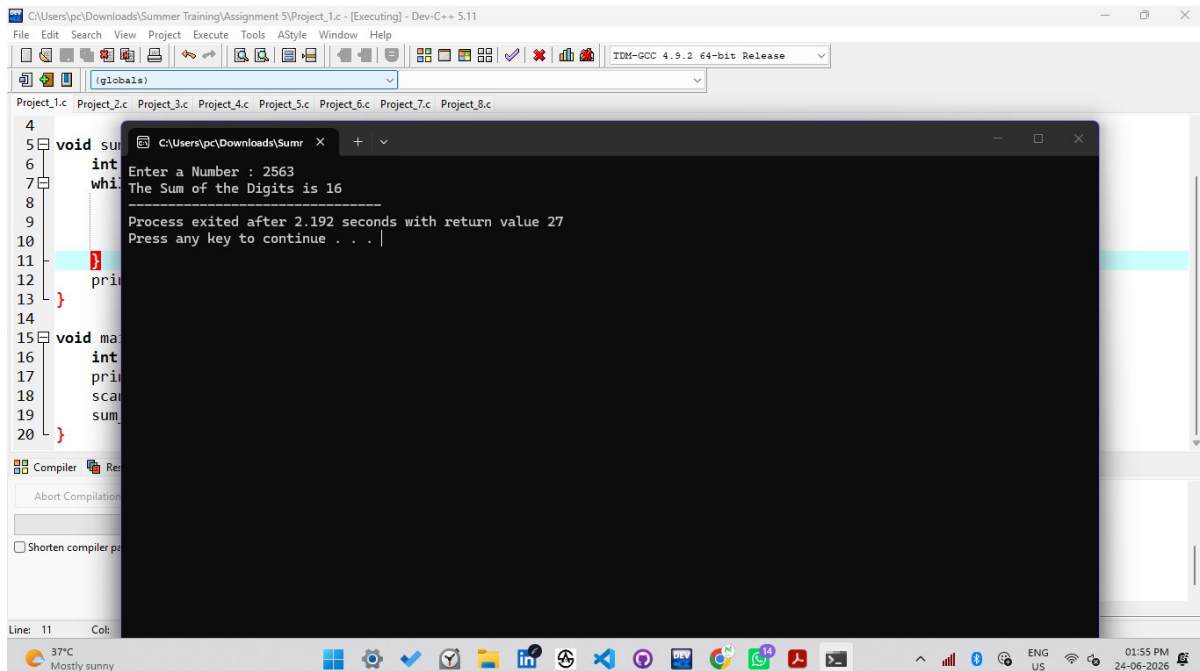
```
// Task 5.1 WAP in C to find sum of digits of a number by using function

#include<stdio.h>

void sum_of_digits(int a){
    int rem, sum = 0;
    while(a > 0){
        rem = a % 10;
        sum += rem;
        a /= 10;
    }
    printf("The Sum of the Digits is %d", sum);
}

void main(){
    int a;
    printf("Enter a Number : ");
    scanf("%d", &a);
    sum_of_digits(a);
}
```

Output



Project 2 — Task 5.2

WAP in C to find factorial of a given number using function

Source Code

```
// Task 5.2 WAP in C to find factorial of a given number using function

#include <stdio.h>

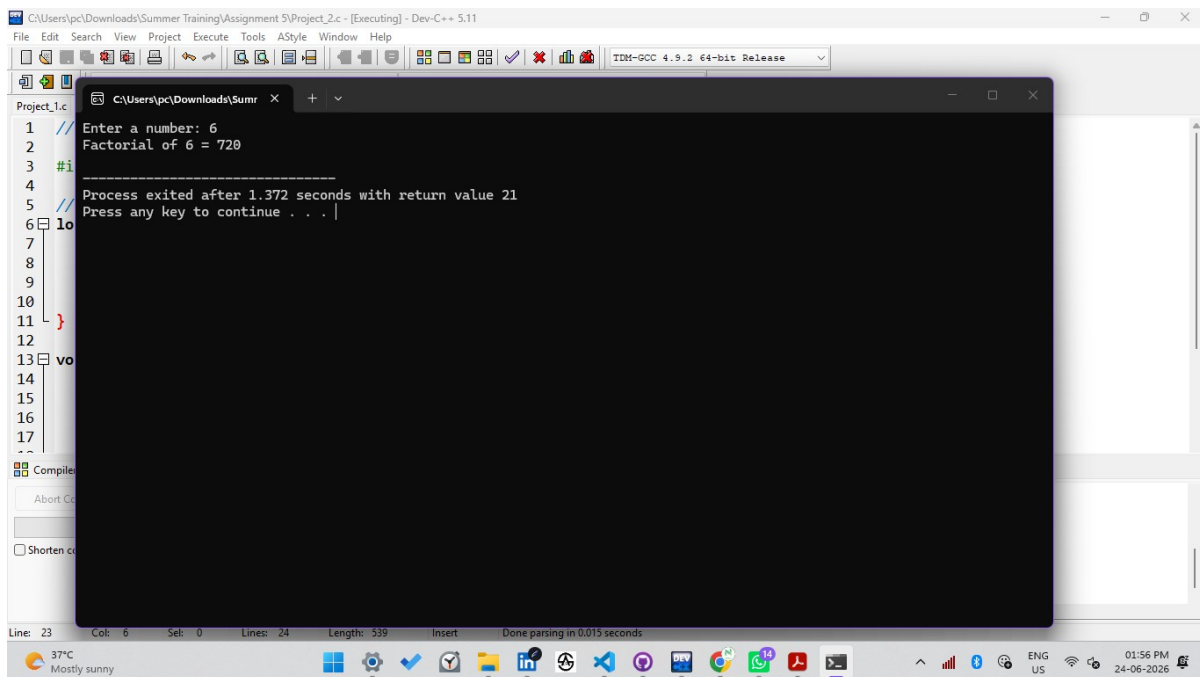
// Function to calculate factorial
long long factorial(int n) {
    if (n == 0 || n == 1)
        return 1;
    else
        return n * factorial(n - 1);
}

void main() {
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if (num < 0) {
        printf("Factorial is not defined for negative numbers.\n");
    } else {
        printf("Factorial of %d = %lld\n", num, factorial(num));
    }
}
```

Output



The screenshot shows the Dev-C++ IDE with a C program being executed. The program prompts the user to enter a number, and the user has entered 6. The output displays the factorial of 6 as 720. The IDE interface includes a menu bar, a toolbar, a file explorer on the left, and a status bar at the bottom showing system information like temperature and time.

```
1 // Enter a number: 6
2 Factorial of 6 = 720
3
4 -----
5 // Process exited after 1.372 seconds with return value 21
6 // Press any key to continue . . . |
7
8
9
10
11
12
13 vo
14
15
16
17
18
```

Project 3 — Task 5.3

WAP in C to find reverse of a given number by using function

Source Code

// Task 5.3 WAP in C to find reverse of a given number by using function.

```
#include <stdio.h>

// Function to reverse a number
int reverse(int n) {
    int rev = 0;
    while (n != 0) {
        int digit = n % 10;
        rev = rev * 10 + digit;
        n = n / 10;
    }
    return rev;
}

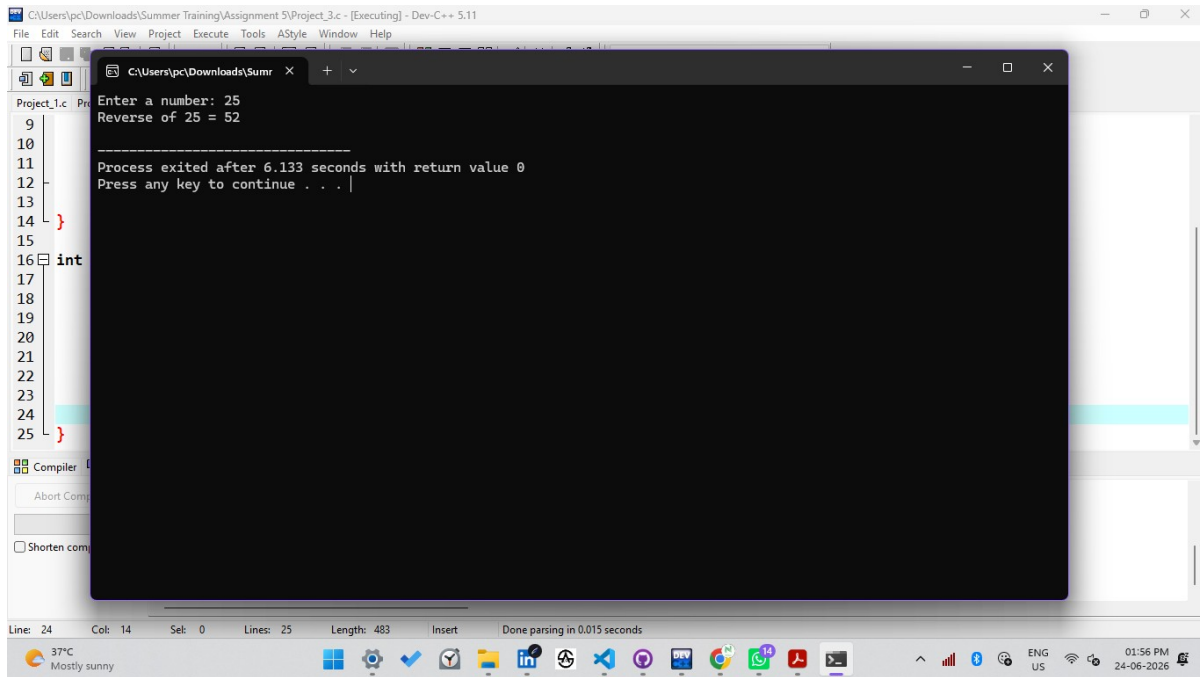
int main() {
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    printf("Reverse of %d = %d\n", num, reverse(num));

    return 0;
}
```

Output



Project 4 — Task 5.4

WAP in C to check given character is vowel or consonant

Source Code

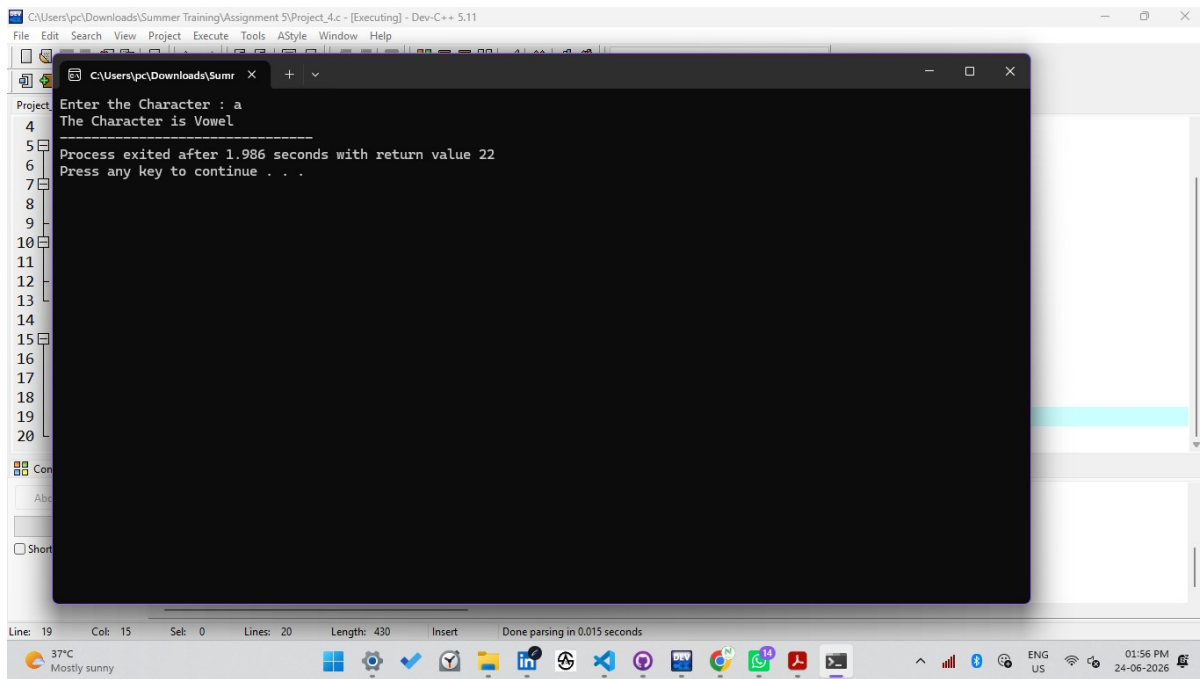
// Task 5.4 WAP in C to check given character is vowel or consonant.

```
#include<stdio.h>

void Check(char ch){
    if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u' ||
       ch=='A' || ch=='E' || ch=='I' || ch=='O' || ch=='U'){
        printf("The Character is Vowel");
    }
    else{
        printf("The Character is Consonant");
    }
}

void main(){
    char ch;
    printf("Enter the Character : ");
    scanf("%c", &ch);
    Check(ch);
}
```

Output



```
Enter the Character : a
The Character is Vowel
Process exited after 1.986 seconds with return value 22
Press any key to continue . . .
```

Project 5 — Task 5.5

WAP in C to change lower case characters of a string to upper case by using function

Source Code

// Task 5.5 WAP in c to change lower case characters of a string to upper case by using function.

```
#include <stdio.h>
```

```
// Function to convert lowercase string to uppercase
```

```
void toUpperCase(char str[]) {
    int i = 0;
    while (str[i] != '\0') {
        if (str[i] >= 'a' && str[i] <= 'z') {
            str[i] = str[i] - 'a' + 'A';
        }
        i++;
    }
}
```

```
void main() {
    char str[100];

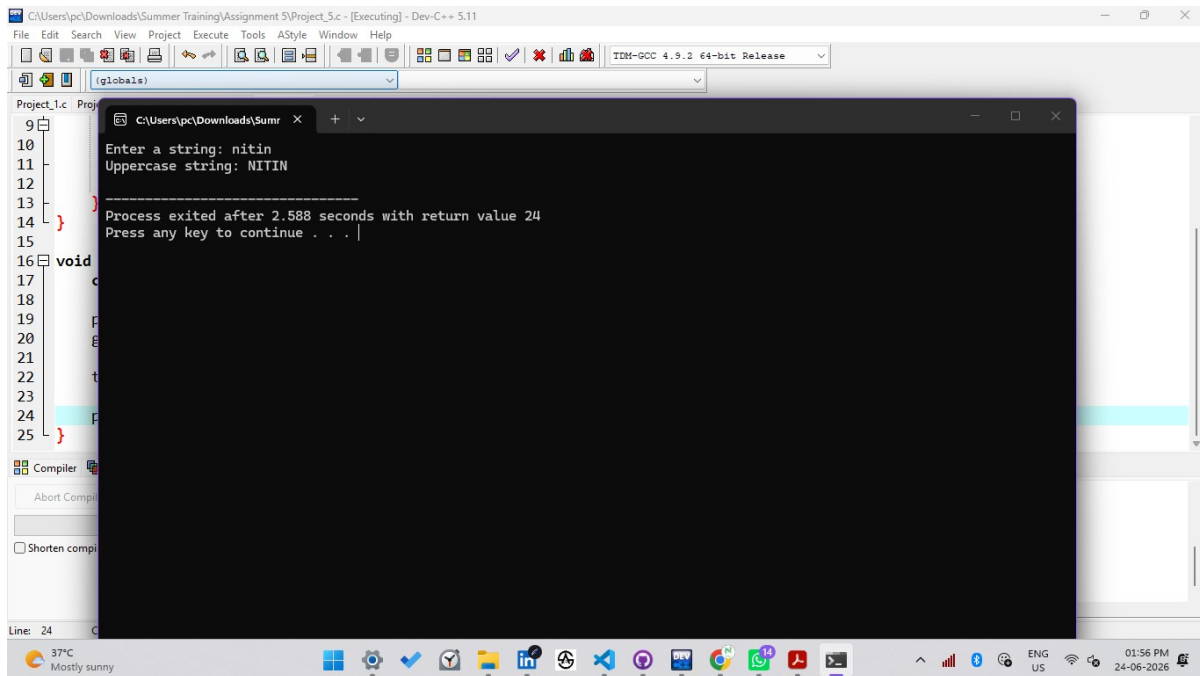
    printf("Enter a string: ");
    gets(str); // or use fgets(str, sizeof(str), stdin) for safety

    toUpperCase(str);

    printf("Uppercase string: %s\n", str);
}
```

}

Output



```
Enter a string: nitin
Uppercase string: NITIN

-----
Process exited after 2.588 seconds with return value 24
Press any key to continue . . .
```

Project 6 — Task 5.6

WAP in C to change upper case characters of a string to lower case by using function

Source Code

```
// Task 5.6 WAP in c to change upper case characters of a string to lower case by using function.
```

```
#include <stdio.h>
#include <string.h>

// Function to convert uppercase string to lowercase
void toLowerCase(char str[]) {
    int i = 0;
    while (str[i] != '\0') {
        if (str[i] >= 'A' && str[i] <= 'Z') {
            str[i] = str[i] - 'A' + 'a';
        }
        i++;
    }
}

void main() {
    char str[100];

    printf("Enter a string: ");
```

```

fgets(str, sizeof(str), stdin);

// Remove trailing newline character left by fgets
str[strcspn(str, "\n")] = '\0';

toLowerCase(str);

printf("Lowercase string: %s\n", str);
}

```

Output

```

C:\Users\pc\Downloads\Summer Training\Assignment 5\Project_6.c - [Executing] - Dev-C++ 5.11
Enter a string: KUMAr
Lowercase string: kumar

Process exited after 11.81 seconds with return value 24
Press any key to continue . . .

```

Project 7 — Task 5.7

WAP in C to find factorial of given number by using recursion

Source Code

```

// Task 5.7 WAP in c to find factorial of given number by using recursion.

#include <stdio.h>

// Recursive function to find factorial
long long factorial(int n) {
    if (n == 0 || n == 1)
        return 1;
    else
        return n * factorial(n - 1);
}

void main() {

```



```

int num;

printf("Enter a number: ");
scanf("%d", &num);

if (num < 0) {
    printf("Factorial is not defined for negative numbers.\n");
} else {
    printf("Factorial of %d = %lld\n", num, factorial(num));
}
}

```

Output

```

C:\Users\pc\Downloads\Summer Training\Assignment 5\Project_7.c - [Executing] - Dev-C++ 5.11
Enter a number: 7
Factorial of 7 = 5040

Process exited after 4.501 seconds with return value 22
Press any key to continue . . .

```

Project 8 — Task 5.8

WAP in C to check given number is prime or not

Source Code

```

// Task 5.8 WAP in c to check given number is prime or not.

#include <stdio.h>

int isPrime(int num) {
    int i;
    if (num <= 1)
        return 0;
    for (i = 2; i <= num / 2; i++) {
        if (num % i == 0)
            return 0;
    }
}

```

```

    }
    return 1;
}

int main() {
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);
    if (isPrime(num))
        printf("%d is a Prime number.\n", num);
    else
        printf("%d is not a Prime number.\n", num);
    return 0;
}
```

Output

